TITLE:- Simulation of Mobile Charger Circuit for Stable 5V Output Using Multisim

Roll No: 12, 36, 35, 38, 37
Branch / Semester: Chemical

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Submitted to:- ANIKET SIR
Subject: IEEE
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INTRODUCTION (1)

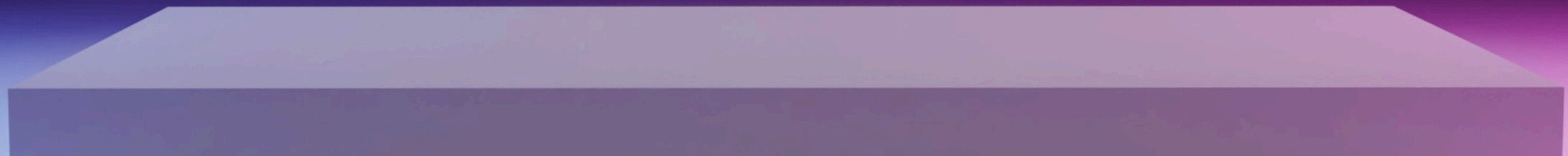
- Mobile phones are essential electronic devices
- Mobile batteries require 5V DC supply
- 230V AC is unsafe for mobile charging
- Charger converts AC into safe DC
- Project focuses on simulation of charger circuit

INTRODUCTION (2)

- Mobile charger is a regulated power supply
- Provides constant output voltage
- Protects from voltage fluctuations
- Uses IC 7805 for regulated 5V
- Simulation avoids hardware damage

PROJECT PROBLEM STATEMENT

- Direct AC supply cannot be used
- High voltage damages batteries
- Voltage fluctuation affects charging
- Need stable, safe 5V DC output



PROJECT OBJECTIVES

- Design mobile charger circuit

- Study transformer, rectifier, filter, regulator

- Simulate circuit using Multisim

- Verify stable 5V DC output

BLOCK DIAGRAM

AC → Transformer → Rectifier → Filter → Regulator → 5V DC

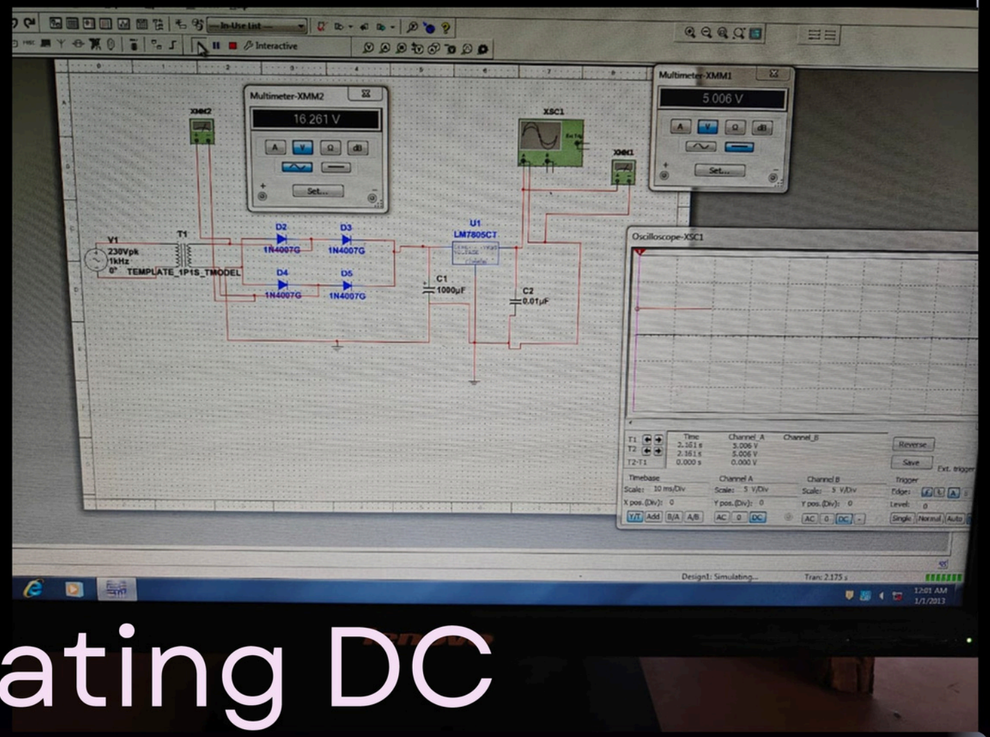
COMPONENTS USED

- 230V/12V Transformer
- 1N4007 Diodes (4)
- 1000µF Capacitor
- IC 7805
- 0.01µF Capacitor

WORKING OF TRANSFORMER

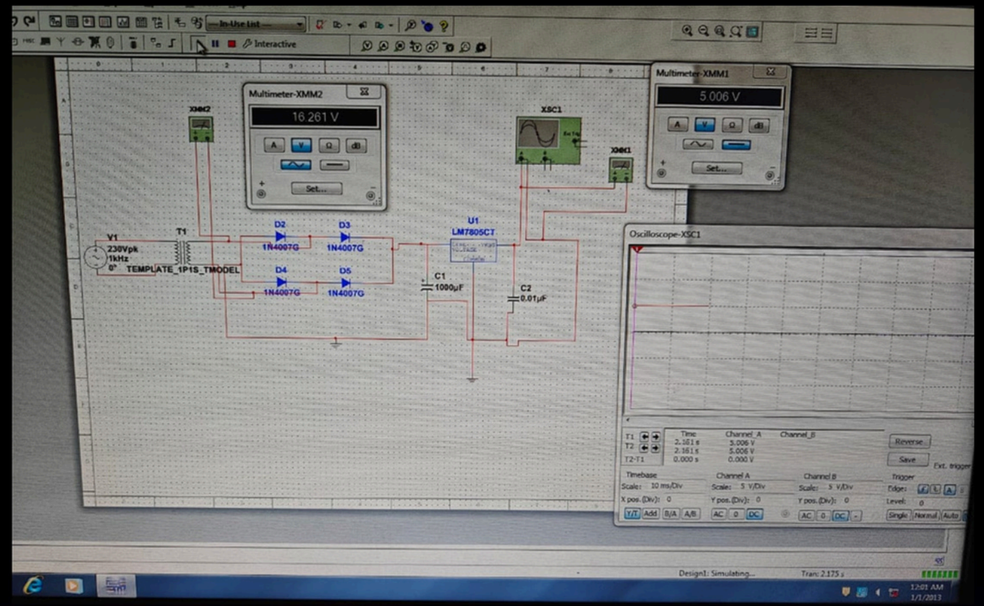
- Electromagnetic induction
- 230V AC → 12V AC
- Provides isolation & safety

WORKING OF RECTIFIER



- Bridge rectifier
- AC converted to pulsating DC
- Unidirectional output

FILTER CIRCUIT



- Capacitor smoothens DC
- Reduces ripple voltag

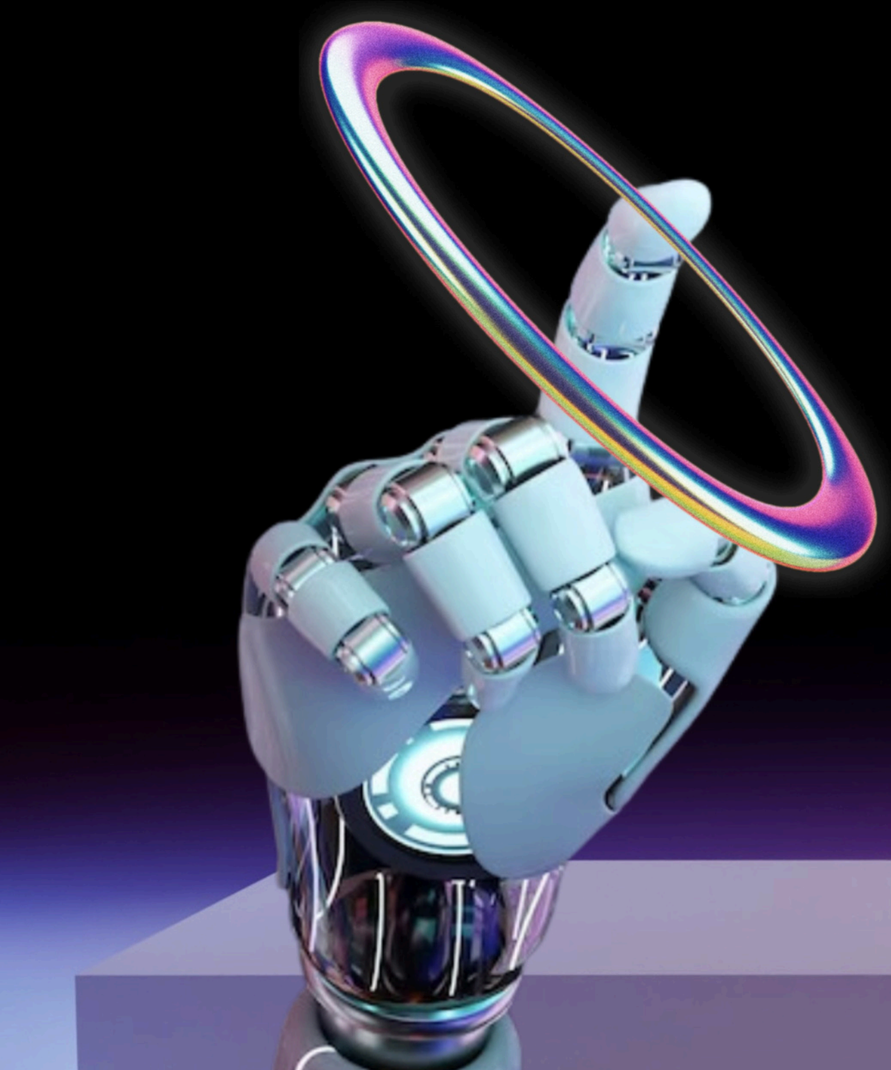
VOLTAGE REGULATOR (7805)

- Fixed 5V regulator
- Handles input/load variation
- Thermal & short-circuit protection

PIN CONFIGURATION (7805)

- Pin 1: Input
- Pin 2: Ground
- Pin 3: Output (5V)

SIMULATION SOFTWARE



- Multisim software used
- Safe virtual testing
- Saves cost & time

SIMULATION PROCEDURE

- Select components
- Connect circuit
- Run simulation
- Observe output

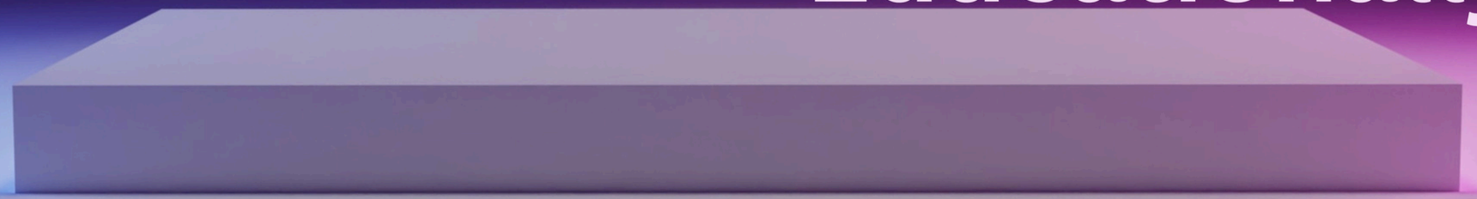


SIMULATION RESULTS

- Input DC ≈ 16V
- Output DC ≈ 5V
- Stable output achieved

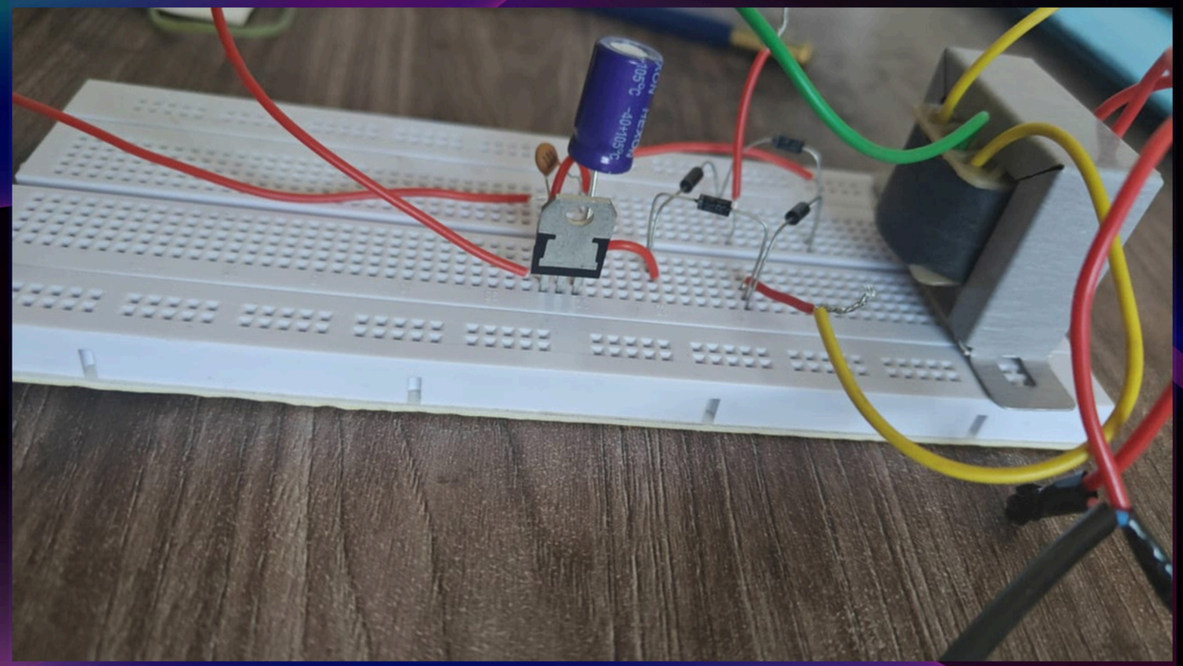
IMPLEMENTATION & RESULTS

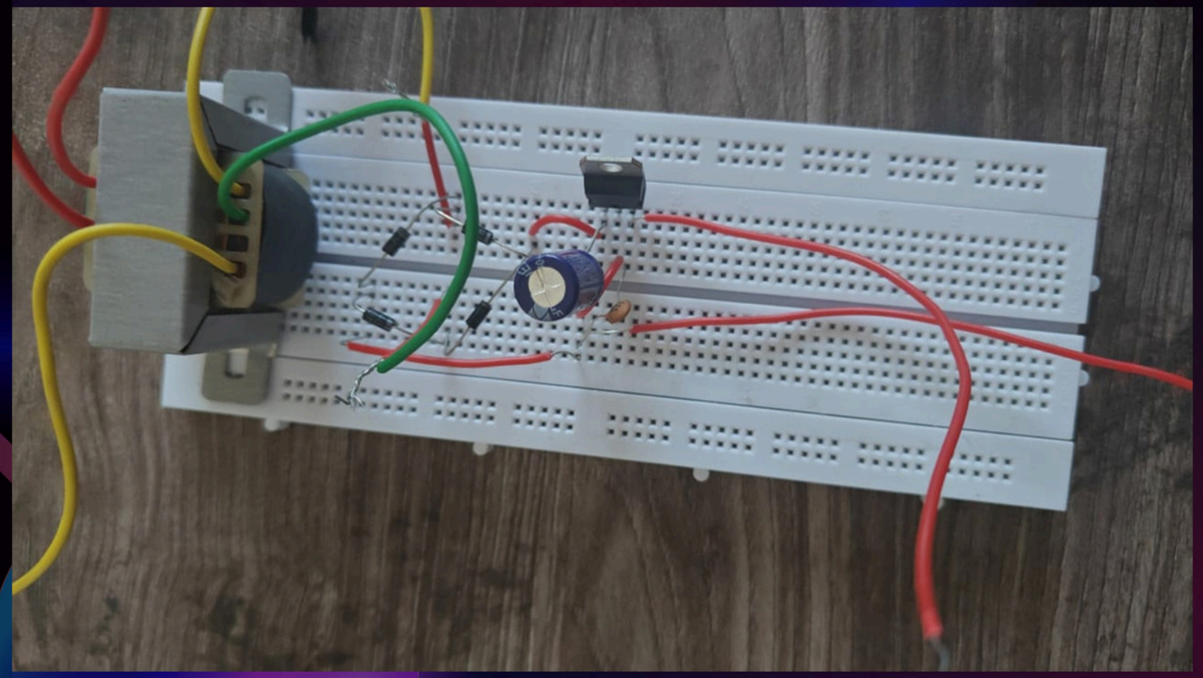
- Circuit successfully simulated
- Stable regulated output
- Educationally useful



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Hardware of mobile charger of 5v output





CONCLUSION & FUTURE SCOPE

Conclusion:- Stable 5V obtained using IC 7805

Future Scope:

- SMPS chargers
- Fast charging
- Higher efficiency

THANK YOU!

Student Teamwork:

- Circuit Design
- Simulation
- Documentation
- Presentation